

Applications and limitations of distributed acoustic sensing in shallow seismic surveys and monitoring

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ABSTRACT

Distributed acoustic sensing (DAS) is an emerging technology gaining acceptance in a variety of seismological applications. We systematically analyze the adaptability and usefulness of horizontal DAS deployments for near-surface geophysical applications, such as surface wave inversion, diving wave tomography, and passive subsurface source location. We find that, in accordance with previous studies, DAS data are generally similar to traditional sensors and can be successfully used independently. Nonetheless, DAS data suffer from inherent limitations due to the design of the optical measurement system. Among others, we identify the gauge length, measurement directivity, and saturation at near offsets as the primary limiting factors. When operating in low-velocity environments and a standard 10 m gauge length, surface wave analysis may be constrained to the usage of low

frequencies due to the wavenumber filtering effect of the gauge length. The measurement directivity generally prohibits applications that are based on upgoing P-wave energy, such as near-off-set diving wave tomography. In addition, saturation at near offsets prevents reliable diving wave traveltime picking. As a consequence of these limitations, the ultrashallow resolution achievable with DAS data is poor. Source directivity also strongly limits passive source location resolution. Whereas some of the limitations can be alleviated through an optimal choice of optical parameters at the interrogator level, we mostly rely on adapted field acquisition and data processing workflows that use the undisturbed portion of the recorded signal. Eventually, the spatiotemporal resolution and relative ease of long-term deployments turn DAS into a worthwhile option for monitoring and time-lapse scenarios, or when operating in urban environments over existing infrastructure.

INTRODUCTION

Distributed acoustic sensing (DAS) is an emerging technology used for a variety of seismic applications. The DAS method enables precise strain or strain-rate recording along a fiber-optic (FO) cable at high spatial and temporal frequencies. A typical DAS system includes a laser interrogator unit (IU) that is connected to an FO cable and continuously sends laser pulses into it. The IU translates the phase shifts in the Rayleigh-scattered laser light to a dynamic strain or strain rate caused by the interaction with seismic waves at each point along the fiber (Lindsey and Martin, 2021). Over the past few decades, DAS has been extensively used in the oil and gas industry, mainly in downhole configurations. It has proven useful for active and passive sensing, onshore and offshore (Mestayer et al., 2011; Mateeva et al., 2014; Daley et al., 2016; Karrenbach et al., 2017; Lellouch et al., 2020). In addition, DAS has been used for earthquake seismology in down-

hole and surface deployments (Lindsey et al., 2017, 2020; Wang et al., 2018; Lellouch et al., 2019), volcanology (Jousset et al., 2022), and for studies in icy terrains (Walter et al., 2020) and marine environments (Williams et al., 2019).

Near-surface DAS operations usually involve deploying FO cables either directly on the surface, burying them in shallow trenches usually a few meters deep, or installing them in boreholes up to a few tens of meters deep. Alternatively, DAS may leverage existing telecommunication fiber networks, thus avoiding installation efforts. However, data quality can strongly vary due to fiber coupling, especially in conduits (Fang et al., 2020). Near-surface DAS deployments were compared with geophones to study their properties (Papp et al., 2014; Martin et al., 2017; Spikes et al., 2019; Abukrat and Reshef, 2021) and used for a large variety of shallow seismic applications: subsurface characterization and imaging (Bakulin et al., 2017; Abukrat and Reshef, 2021), time-lapse monitoring

Manuscript received by the Editor 10 September 2022; revised manuscript received 19 November 2022; published ahead of production 3 May 2023; published online 21 June 2023.

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(Dou et al., 2017), and surface wave dispersion analysis from distinct sources (Yuan et al., 2020) and seismic interferometry (Dou et al., 2017; Martin et al., 2017).

Using DAS for shallow seismic applications potentially holds several key advantages over conventional sensors. DAS FO cables can be installed over very large distances of up to tens of kilometers, amounting to a lower per-channel cost than traditional recording systems. The fiber also may be deployed in complex geometries that include vertical and horizontal deployments, which are challenging to recreate with traditional point sensors. Barring physical damage to cables, DAS installations are usually useful for years, and potentially decades, making them ideal for time-lapse or other monitoring projects. Finally, using DAS over an existing FO infrastructure is an excellent alternative to conventional sensors in situations where a nonintrusive acquisition is required, such as in urban areas.

However, DAS might face some major drawbacks in near-surface seismic applications. First, the mechanical coupling between the fiber and the ground is crucial for the acquisition of high-quality seismic data. Such coupling might degrade due to topography, vegetation, and suboptimal installation. Poor coupling can dramatically compromise the quality of the data. Data quality can be improved by burying the cables in trenches or by coupling them to the ground using heavy objects such as sandbags (Spikes et al., 2019). DAS measurements also might suffer from the effect of the gauge length on the acquired data (Dean et al., 2015). Effectively acting as a spatially moving averaging window, the gauge length causes notches of zero amplitude in the fiber's response when the product of the gauge length by the recorded wavenumber k equals an integer. It also deteriorates the signal when this product is more than approximately 0.2 (Dean et al., 2017). Due to the low-velocity nature of the shallow subsurface, the seismic wavefield is composed of high wavenumbers, and the detrimental effect of the gauge length is expected to be pronounced.

Despite the wide range of DAS near-surface seismic applications, an integrated and comprehensive study dedicated to its usefulness and limitations in the shallow subsurface has not been published yet, and comparing the performance of DAS and traditional 3C geophones in near-surface applications is necessary. In this study, we use 2D active surface data from different types of sources recorded by 3C geophones and a horizontal DAS array to comprehensively compare DAS and traditional geophones. We examine single- and multichannel records and their spectra in multiple acquisition sites. Through the analysis of surface wave dispersion images, we illustrate the potentially detrimental effect of the gauge length on the DAS data. Then, we build a P-wave velocity model of the shallow subsurface using traveltime tomography based on first arrivals picked over DAS data. Finally, we demonstrate how surface DAS records can be used to locate a subsurface source in a monitoring scenario, based on the obtained P-wave velocity model and an existing S-wave velocity model.

DATA ACQUISITION AND EXPERIMENTAL SETUP

We use standard, single-mode FO cables, connected to a Silixa iDAS IU to acquire the data discussed in this study. We use this IU to measure the strain rate along the fiber using a gauge length of 10 m, 1 m channel spacing, and a sampling rate of 1 kHz. The DAS arrays were deployed in horizontal configurations, in which the FO cables are buried inside 0.8 m deep surface trenches. Here,

5 Hz 3C geophones were positioned on the surface with a receiver spacing of 1 or 2 m (differs by experiment) and sampled at 1 kHz, the same as for the DAS system. Most of the data recorded by these two types of sensors were acquired using a time-zero trigger. However, due to technical issues, some recordings are in a continuous mode, and no source origin time information is available. The clocks in both acquisition systems are synchronized through global positioning system.

Most of the data presented in this paper were acquired in a test site referred to as area A, characterized by a sandy-shale compaction-dominated structure with relatively low seismic velocities. A small part of the data was acquired in a geologically different test site, called area B, which is dominated by limestones and dolomites, resulting in relatively high seismic velocities.

COMPARISON BETWEEN DAS AND 3C GEOPHONES

Measurement conversion

Converting DAS strain or strain rate to particle velocity or acceleration can be done by using a semblance-based slant-stack method (Lellouch et al., 2019) or a frequency-wavenumber (f - k) method (Lior et al., 2021). As previously mentioned, our DAS IU measures the strain rate along the direction of the fiber. Assuming a horizontal fiber deployed along the x -axis, the strain rate is given by

$$\dot{\epsilon}_{xx} = \frac{\partial}{\partial t} \left(\frac{\partial u_x}{\partial x} \right), \quad (1)$$

where u_x is the x -component of the displacement field. In contrast, a 3C geophone measures the particle velocity along three orthogonal directions (x -, y -, and z -axes):

$$\dot{\mathbf{u}} = \left(\frac{\partial u_x}{\partial t}, \frac{\partial u_y}{\partial t}, \frac{\partial u_z}{\partial t} \right). \quad (2)$$

To convert DAS strain-rate data to its particle velocity equivalent, we spatially integrate DAS records along the direction of the fiber:

$$\int \frac{\partial}{\partial t} \left(\frac{\partial u_x}{\partial x} \right) dx = \frac{\partial u_x}{\partial t}. \quad (3)$$

Multichannel comparison

We first show multichannel comparisons of DAS and 3C geophone data from two shot gathers acquired at area A, using two different sources. The survey acquisition geometry is identical for both sources and is summarized in Figure 1. The source line consists of 145 surface shots at 1 m intervals along the x -axis. The DAS array and geophone spread include 81 channels that are spaced by 1 m for a total of 80 m along the x -axis. The first source is a vertical impact of a 5 kg sledgehammer on a steel plate, excited at the end of the source line (i.e., shot 145). Converted DAS and 3C geophone seismograms are shown, along with their spectra, in Figure 2. The DAS data suffer from high-frequency noise that is less prominent on the geophones' x - and y -components and barely noticeable on the geophone's z -component. However, some of the

geophone traces, especially in the x - and y -components, are incoherent. In addition, when we processed the data, we realized that three of the geophones were “dead” and did not record any signal. Clear first arrivals of P and S waves are evident on the DAS record and in the geophones’ x - and z -components, whereas only an S arrival can be identified on the y record. The attenuation of frequencies as we obtain farther from the source is observed in all records except the geophone’s y -component. Overall, DAS data are relatively similar to that of the geophone’s x -component in the temporal and frequency domains and to that of the geophone’s z -component in the frequency domain.

For the second source, we replaced the hammer with a horizontal electromagnetic vibrating source (Brodic et al., 2019) oriented in the y -direction (i.e., in the crossline direction). The vibrating source was operated using a sweep of 10–100 Hz at the same shot location as for the previous example. We show converted DAS and 3C geophone seismograms after crosscorrelation with the sweep, along with their spectra, in Figure 3. The P-wave arrivals are much weaker and less clear than for the sledgehammer shot, and can only be seen on the DAS and geophone x and z records. Strong arrivals of S waves are evident in the DAS and geophone x and y data, but much less obvious in the z -component data. Moreover, the high-frequency part of the vibroseis sweep, which we interpret as a surface wave, arrives at approximately 350 ms in the DAS and in all geophone components. It is visible after the low-frequency surface waves and its dissipation is clear due to its high frequency. In areas in which the velocity increases with depth, surface waves are dispersive, with the low frequencies propagating faster. The back scattering at approximately 550 ms that originates around channel 60 might be caused by a $1\text{ m} \times 1\text{ m} \times 1\text{ m}$ nearby concrete structure. Such scattering (diffraction) is expected only at high frequencies (i.e., short wavelengths), thus reinforcing our assumption that these high-frequency arrivals are meaningful signals. However, they are not properly recorded by the DAS system. They are distorted and suffer from discontinuities and irregularities that are not observed in any of the geophone components. We attribute this to the DAS gauge length. The effect of the gauge length is equivalent to applying a wavenumber-dependent filter on the data of the form $(\sin(\pi kGL))/\pi k$, where k is the wavenumber and GL is the gauge length (Dean et al., 2015, 2017), resulting in a significantly altered DAS seismogram compared with the geophone data. Seismic phases with short wavelengths, or high temporal frequencies, are strongly attenuated due to the gauge length. From a spectral point of view, the attenuation of frequencies is observed in DAS and geophones, and the frequency content of the DAS data is significantly lower than for all three geophone components, as predicted.

Single-channel comparison

Spatially integrated DAS strain-rate records are theoretically equivalent to the x -component of the particle velocity measured by the geophones. In this section, we compare individual DAS channels with geophone x -components, shown to be the most resemblant to DAS data. Figure 4 shows seismograms and spectra, both after conversion, of the hammer shot ($x = 144\text{ m}$) shown in Figure 2 for two single traces — the first receiver of the line ($x = 0\text{ m}$) and the middle receiver of the line ($x = 40\text{ m}$). The seismograms are significantly closer for the larger source-receiver offset ($x = 0\text{ m}$), showing similar behavior. Due to the attenuative nature of the shallow subsurface, the dominant frequency of these data is much lower than for the receiver located at $x = 40\text{ m}$. The DAS signal at $x = 40\text{ m}$ contains much lower frequencies than those observed on the corresponding geophone’s x -component trace, indicating the detrimental effect of the gauge length on the high-frequency components of the data.

Figure 5 shows the seismograms and spectra after conversion of the electromagnetic vibrator’s sweep from Figure 3 ($x = 144\text{ m}$) for the same two single traces (located at $x = 0\text{ m}$ and $x = 40\text{ m}$). Because we did not observe relatively flat spectral energy within the source operating band of 10–100 Hz, as expected for a vibrational source, we first validated that there were no instrumental issues by examining the spectra of adjacent geophones, which were consistent. A possible explanation is that the coupling between the electromagnetic vibrator and the ground was insufficient and varied as a function of frequency. In terms of recorded data, the behavior is quite similar to the hammer shots, but for the receiver at $x = 40\text{ m}$ (Figure 5a), the DAS signal significantly differs from the geophone. The attenuation of the higher frequencies from the middle-receiver location ($x = 40\text{ m}$) to the first-receiver location ($x = 0\text{ m}$) is noticeable in both sensors. However, even at this station, the DAS frequency content is narrower than for the geophone. In addition, the DAS trace at $x = 0\text{ m}$ suffers from a high-frequency noise of approximately 90 Hz. The geophone’s x -component trace located at the middle receiver also suffers from a high-frequency noise of approximately 95 Hz. We could not identify the source of these noises.

ANALYSIS OF DISPERSION IMAGES

In this section, we analyze the dispersion images of DAS data. Our focus is to further examine the gauge length effect on the DAS data and to quantify its effect on surface wave inversion, a widespread procedure in near-surface geophysics (Yilmaz, 2021). Dispersion images show an amplitude map of phase velocity versus frequency (Park et al., 1999, 2007). If the wavenumber k is equal to $(1/GL) = 0.1\text{ m}^{-1}$, the DAS recorded amplitude should be zero.

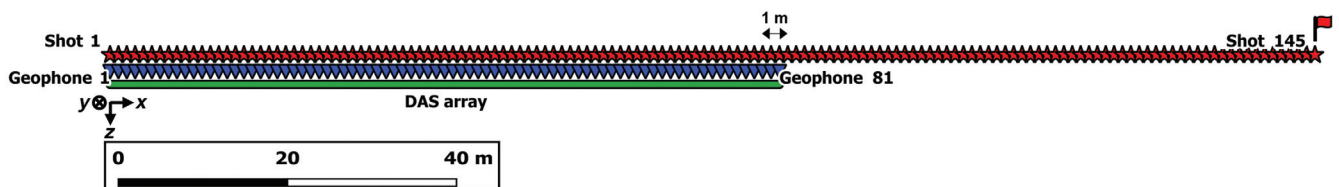


Figure 1. Survey acquisition geometry. The shots are denoted by the red stars and the geophones are denoted by the blue triangles. Both are spaced by 1 m, for a total of 145 shots over 144 m and 81 geophones over 80 m. The green line represents the DAS array, which begins at the location of the first shot and receiver ($x = 0\text{ m}$) and ends at the location of the last geophone ($x = 80\text{ m}$). The red flag ($x = 144\text{ m}$) indicates the location of the hammer and vibroseis shots presented in Figures 2 and 3, respectively.

Above that value, the amplitude and phase of the recorded signal will be heavily distorted (Dean et al., 2015, 2017). Therefore, over the frequency range of interest, we can calculate a threshold line of the form $v = (f/0.1)$, below which we expect to measure only noise.

Figure 6 shows dispersion images from two shot records of the vibrational source recorded by DAS and the x -component of 3C geophones in areas A and B. The shot from area A is the shot shown in Figure 3, and its dispersion images (Figure 6a and 6b) exhibit relatively low phase velocities. The seismograms of the shot from area A are shown in Figure 3, and its dispersion images (Figure 6a

and 6b) exhibit relatively low phase velocities. In Figure 6a, on the DAS dispersion image from area A, there are no events below the $k = 0.1 \text{ m}^{-1}$ line, whereas, for the geophone's x -component (Figure 6b), there is a clear, high-amplitude fundamental mode that is coherent even at high frequencies. For this low-velocity area, the gauge length strongly limits the resolution of surface wave inversion methods, often used with DAS data (Dou et al., 2017; Martin et al., 2017). In contrast, the dispersion images of both sensors from area B look almost identical. The effect of the gauge length seems to be insignificant due to the high phase velocities combined with the relatively low-frequency content of the data.

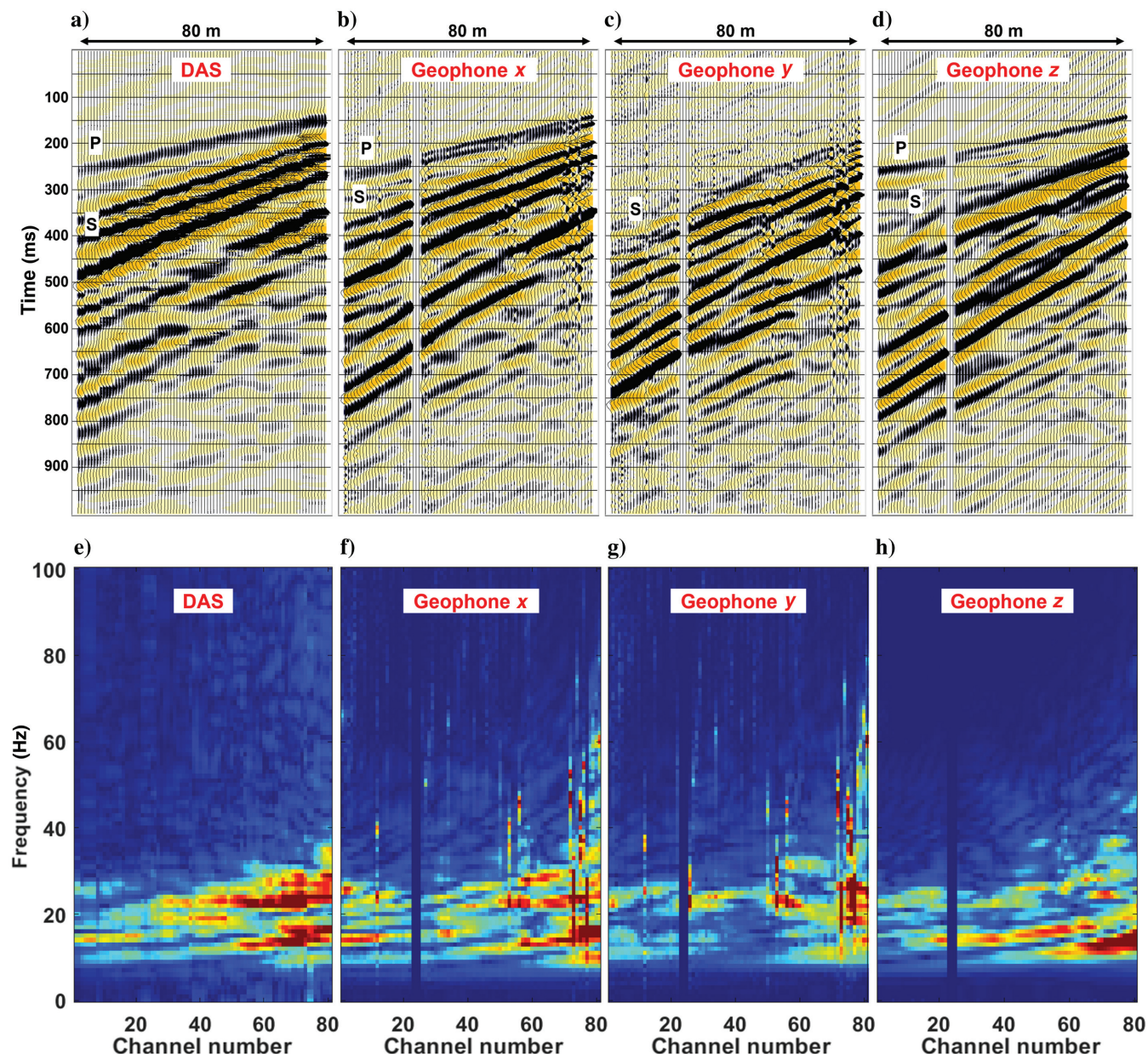


Figure 2. Multichannel comparison of seismograms from a hammer shot: (a) DAS, (b) geophone's x -component, (c) geophone's y -component, and (d) geophone's z -component. The first arrivals of the P and S waves are marked. The shot location is indicated by the red flag in Figure 1 ($x = 144 \text{ m}$). Power spectra comparison of the hammer shot seismograms: (e) DAS, (f) geophone's x -component, (g) geophone's y -component, and (h) geophone's z -component. Each trace is individually scaled, and higher power values are represented by warmer colors.

DIVING WAVE TRAVELTIME TOMOGRAPHY

Seismic diving wave tomography is a commonly used velocity estimation technique that uses first-arrival traveltimes to estimate the velocity structure of the subsurface. The subsurface is represented by either a fixed- or variable-sized grid. Then, using forward-modeling techniques, raypaths and traveltimes between source-receiver pairs are calculated. The velocities in each grid cell are iteratively updated until the misfit between the observed and calculated traveltimes reaches some threshold value (White,

1989; Sheehan et al., 2005). This method requires picking, either automatically or manually, the first arrivals in each seismogram.

To assess the feasibility of DAS for shallow velocity-model building, we conducted a 2D surface survey at area A. Figure 7a shows the acquisition setup. The 121 hammer shots are spaced 1 m apart for a total of 120 m along the x -axis. The DAS array contains 211 channels at 1 m intervals for a total of 210 m, whereas the geophone spread, due to logistical issues, only consists of 50 channels at 2 m intervals for a total of 98 m, both along the x -axis. Figure 7b–7d shows the manually picked first arrivals overlaying the DAS seismograms, after

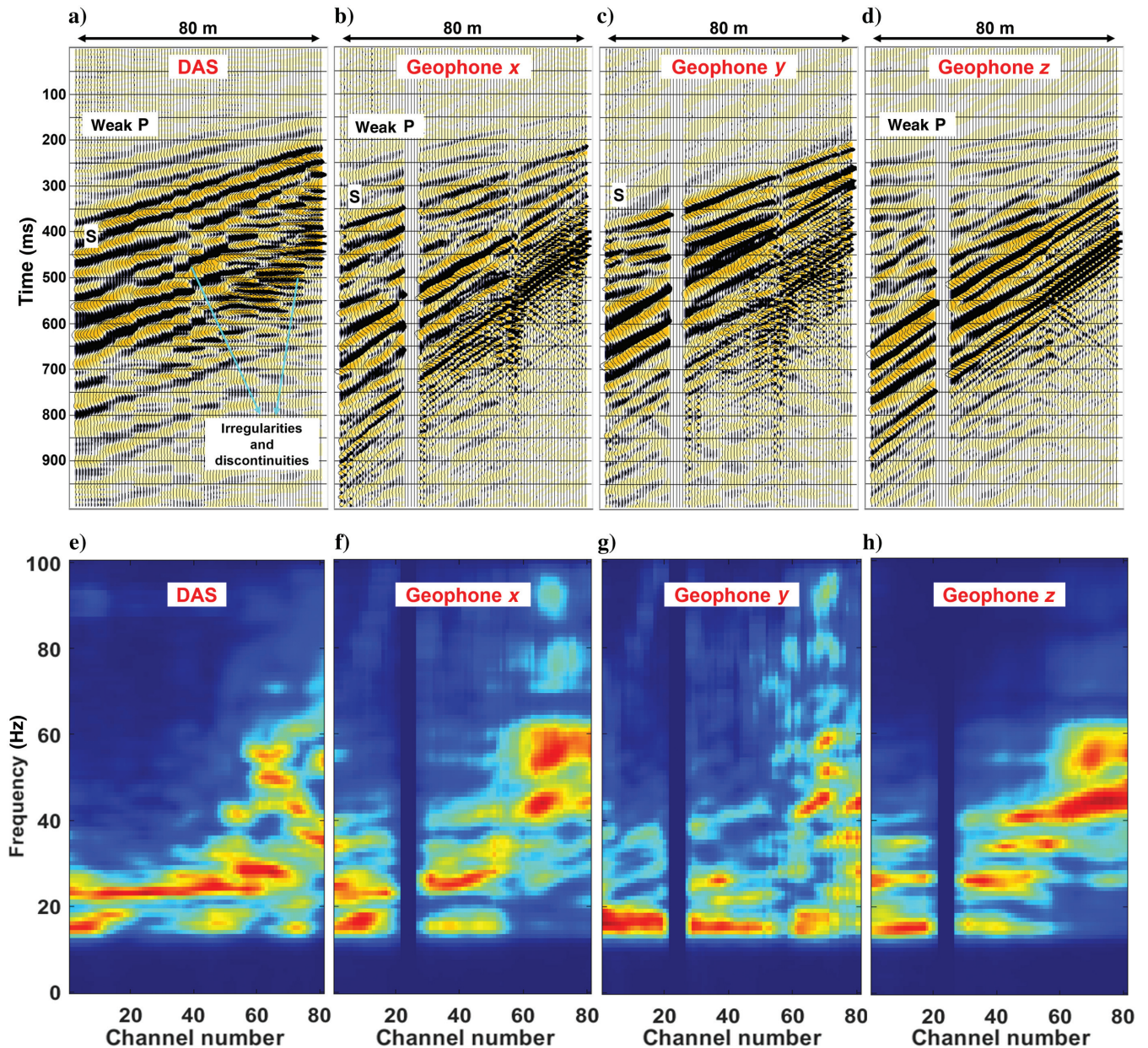


Figure 3. Multichannel comparison of correlated seismograms from a vibroseis sweep: (a) DAS, (b) geophone's x -component, (c) geophone's y -component, and (d) geophone's z -component. Examples of irregularities and discontinuities as well as the first arrivals of P and S waves are marked. The sweep location is indicated by the red flag in Figure 1 ($x = 144$ m). Power spectra comparison of the vibroseis seismograms after correlation: (e) DAS, (f) geophone's x -component, (g) geophone's y -component, and (h) geophone's z -component. Each trace is individually scaled, and higher power values are represented by warmer colors.

the conversion to particle velocity, for different source locations. During the manual picking of the first arrivals, we ignored the noisy near-offset traces and some low-confidence traces. The near-offset channels are highly disturbed and incoherent because they are saturated due to strain values that exceed the upper limit of the DAS system (Abukrat and Reshef, 2021). Although an increased sampling rate can mitigate this problem, the highest possible sampling rate in our DAS system is automatically chosen for a given fiber length. Because here we use a short section of a much longer cable, it limits the internal sampling rate of the system. A low-frequency noise with an inverse moveout is visible in the last shot of the line (Figure 7d). The source of this noise was a distant vehicle driving outside of the line, and we had no control over it. However, it did not affect the picking procedure. We did not want to apply any filters to the data to avoid phase alteration effects.

Figure 8 shows the resulting P-wave velocity model computed from the tomography using the DAS and geophone picks. When considering the first 120 m of the model which are covered by the source line, the geophone spread, and a segment of the DAS array of the same length as the spread of geophones, the DAS (Figure 8a) provides a less-detailed velocity model with limited resolution compared with the geophone model (Figure 8b). All of the geophone's components were picked together, whereas, for each shot record, we examined all the components simultaneously and picked the one with the highest confidence. The geophone-based model reveals features that are invisible in the DAS model. A shallow low-velocity (approximately 300 m/s) layer is clearly

seen down to a depth of 3 m on the geophone model, whereas it is not detected in the DAS model. A high-velocity zone (approximately between $x = 30$ and $x = 60$) appears focused on the geophone model, whereas it is smeared in the DAS model. We attribute these discrepancies to the fact that we did not pick the near-offset traces on the DAS record as they suffer from a lack of coherency. In addition, the DAS system is less sensitive to arrivals with near-vertical incidence, for which the z -component of the geophone is ideal. Furthermore, Figure 8c combines the velocity models of the geophone spread and the complete 210 m long DAS array. When the calculation includes additional components and more offsets are taken into account, a much more detailed velocity model is obtained. The combined model shows a complex shallow subsurface that would not be obtained by having only one of the sensors.

SUBSURFACE SOURCE LOCATION

The accurate location of shallow subsurface sources is important in monitoring precursors to environmental hazards (such as sink-holes and landslides), underground intrusion detection, rescue operations of trapped personnel, tracking structural integrity of mines, and more. The location method that we use is based on a simple traveltimes error minimization (Pesicek et al., 2014). We start by picking the first-arrival times of the P and S waves, wherever possible. Then, we divide the survey area into a grid of cells and calculate the traveltimes from each cell to all the receiver locations using the eikonal equation and known velocity models.

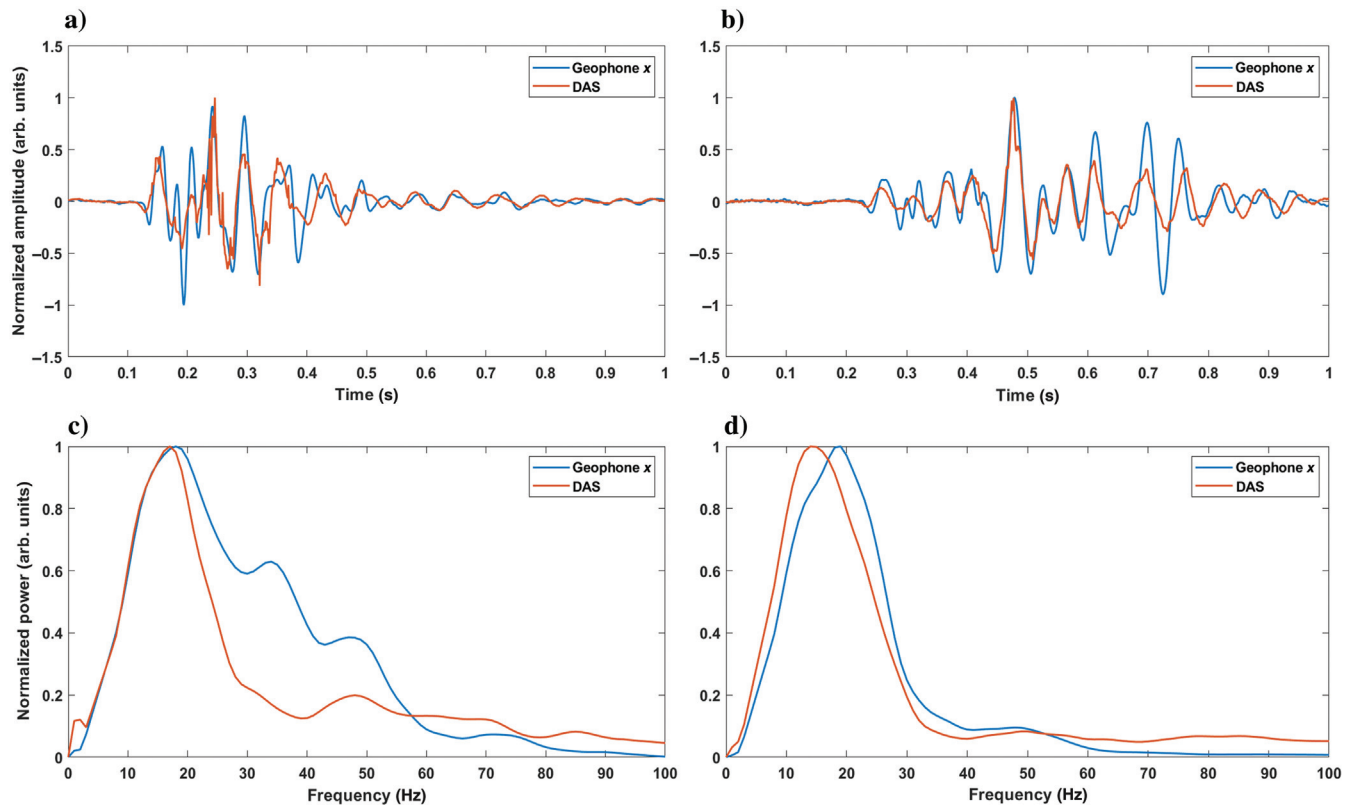


Figure 4. Single-channel comparison of DAS and geophone's x -component seismograms and spectra from the hammer shot ($x = 144$ m) in Figure 2. (a) Seismograms from the middle of the line ($x = 40$ m). (b) Seismograms from the beginning of the line ($x = 0$ m). (c) Corresponding spectra of the traces in (a). (d) Corresponding spectra of the traces in (b). The shot location is indicated by the red flag in Figure 1.

The cell with the minimal misfit between the predicted and measured traveltimes is chosen as the source location.

To compare the functionality of DAS and 3C geophones for locating shallow subsurface sources, we carried out a field experiment in area A, as shown in Figure 9. Data were acquired using a subsurface source, a DAS array, and a spread of 3C geophones. The DAS array and the geophone spread consist of 50 stations at 2 m intervals and have the same total length of 98 m. The subsurface accelerated-mass source is activated at a depth of 15 m with horizontal coordinates of $x = 0$ m and $y = 25$ m, and its orientation is perpendicular to the DAS and geophones.

We show seismograms of DAS after conversion and 3C geophones in Figure 10. The DAS provides data of higher quality, and events can be picked more easily than in all 3C of the geophone. We estimate that DAS is superior in this case because the fiber is buried in a trench whereas the geophones are deployed on the surface, increasing their exposure to noise. We identify the first coherent phase to reach the geophones' z -component as a P wave, and the first coherent phase on the DAS and the geophones' x - and y -components as an S wave. The first reason for this interpretation is the arrival time of the events. At the time of the P-wave arrival, there is nothing but noise on all the seismograms except in the z -component record. Second, as seen on the seismograms, the phases have different moveouts, where what we identify as the P wave is faster than the S wave. Differences in moveouts are also evident in Figure 11, which displays the picked traveltimes in each station for the DAS and geophones. As we do not have the source origin time, the dis-

played traveltimes are relative to the first arrival of the P wave that was picked on the geophone's z -component record. The geophone S picks are an average of the picks from the geophone's x - and y -components as they were quite similar. The DAS S picks are very close to the S picks of the geophone, increasing our certainty that the same phase is picked on both records.

Another interesting property of the DAS data that is not observed in any of the geophone components is the polarity reversal at the apex (see Figure 10). If we assume mild lateral velocity changes, this noticeable polarity reversal on the DAS data could be a horizontal indicator of the subsurface source, similar to Cole et al. (2018).

Figure 12 compares the DAS and 3C geophone location results, obtained using the combined P-wave velocity model shown previously (Figure 8c), an existing S-wave velocity model from previous studies in the area, and the observed P and S picks (Figure 11). The S-wave velocity model that we used was obtained by a crosswell acquisition using downhole 3C geophones and a downhole directional source. However, due to logistical and operational issues during our experiment, we could not acquire DAS data in such a crosswell configuration. Because of that and because crosswell velocity models are much more accurate than velocity models obtained by surface acquisitions (Lellouch and Landa, 2017; Lellouch and Reshef, 2017), we preferred to use a robust existing model. The location using the DAS S picks (Figure 12a) yields a subsurface source location of $x = 4$ m, $y = 31$ m, and a depth of 18 m, compared with the location obtained by the average S picks of the geophone's x - and y -components (Figure 12b) at $x = 5$ m,

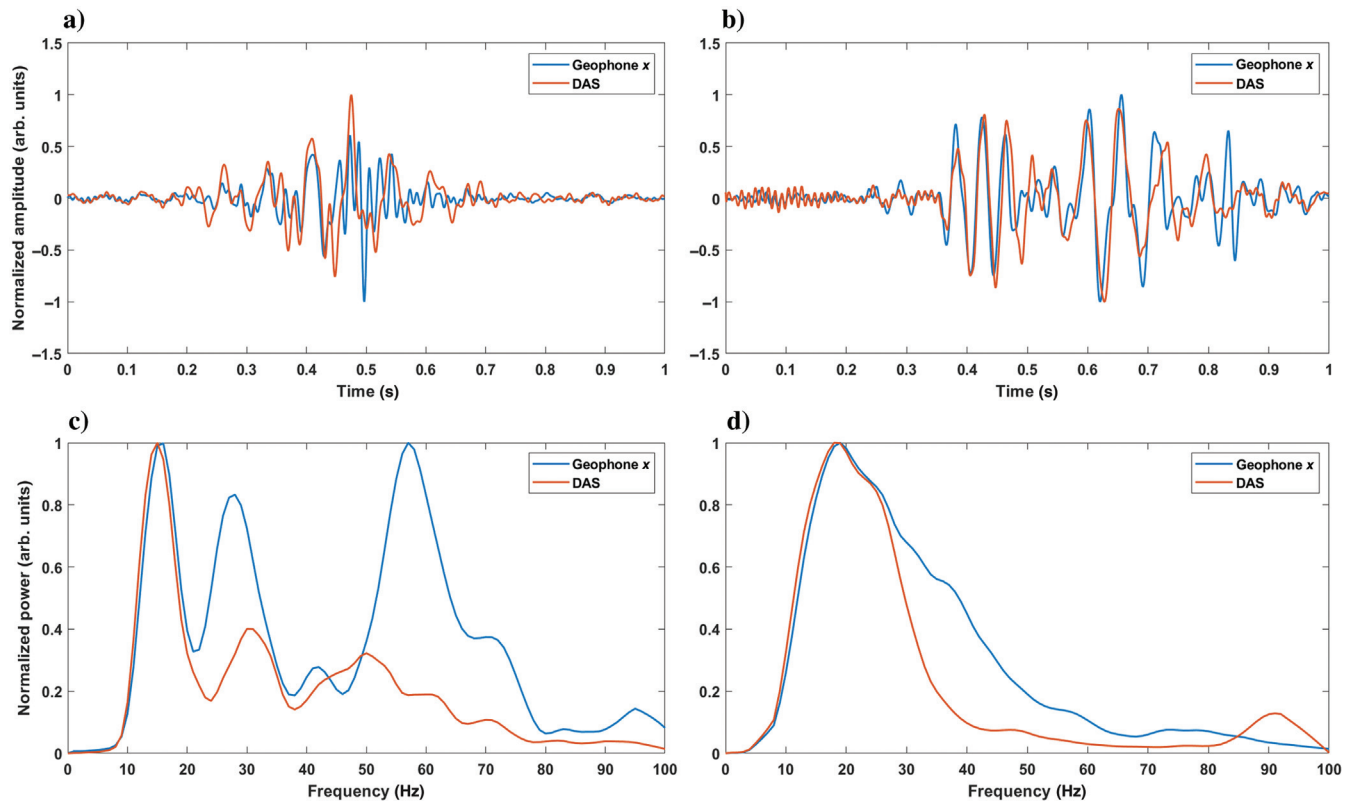


Figure 5. Single-channel comparison of DAS and geophone's x -component seismograms and spectra from the vibroseis sweep ($x = 144$ m) in Figure 3. (a) Seismograms from the middle of the line ($x = 40$ m). (b) Seismograms from the beginning of the line ($x = 0$ m). (c) Corresponding spectra of the traces in (a). (d) Corresponding spectra of the traces in (b). The sweep location is indicated by the red flag in Figure 1.

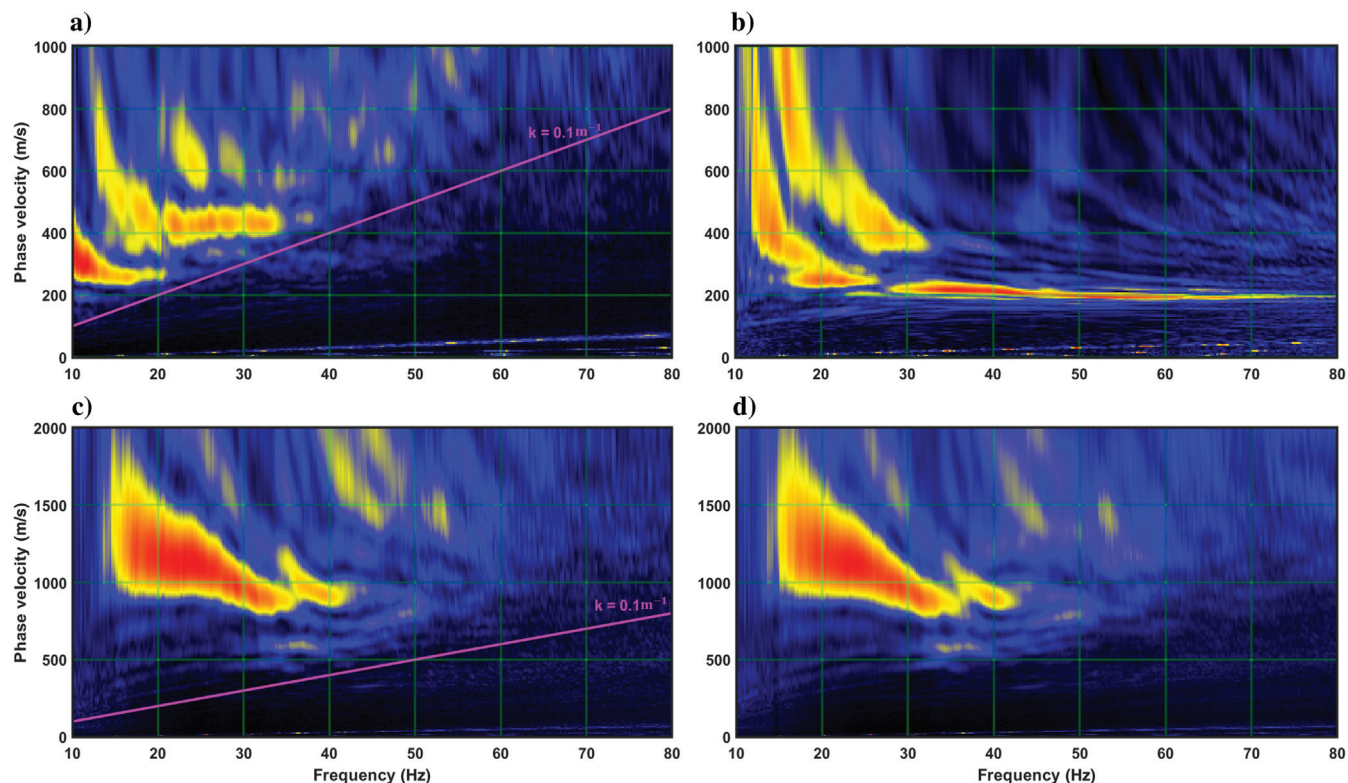


Figure 6. Comparison of DAS and geophone's x -component dispersion images from two distinct areas: (a) DAS from area A, (b) geophone's x -component from area A, (c) DAS from area B, and (d) geophone's x -component from area B. The magenta line in (a and c) represents the gauge length effect threshold curve of $v = (f/0.1)$. Higher amplitude values are represented by warmer colors.

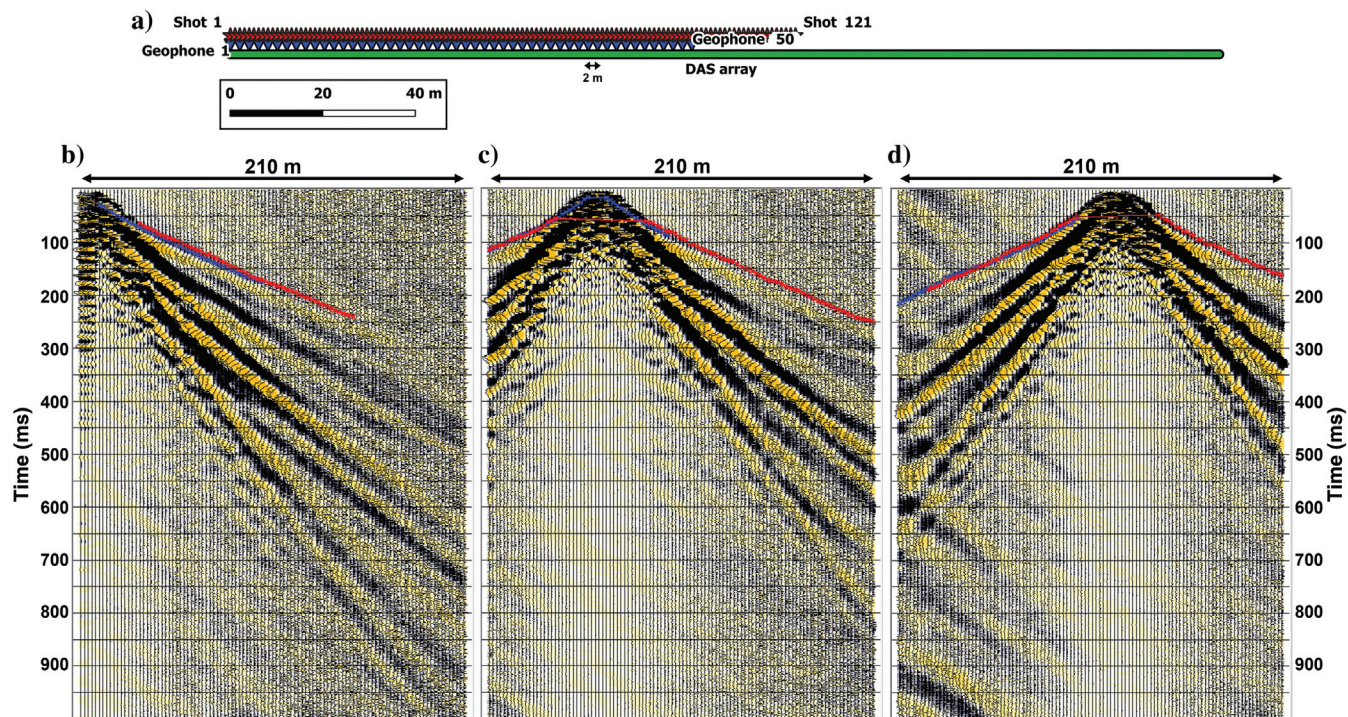


Figure 7. Surface shots for the tomography. (a) Survey acquisition geometry. The shots are denoted by the red stars and spaced by 1 m, the geophones are denoted by the blue triangles and spaced by 2 m, and the DAS array is indicated by the green line. (b–d) Shots acquired by the DAS array from the beginning, middle, and end of the source line, respectively. The picked DAS and geophone first arrivals are indicated by the red and blue lines, respectively.

$y = 30$ m, and a depth of 12 m. For completeness, we computed the location using the S-arrival picks from the geophone's x - and y -components separately, but they were identical to the location estimated using the average picks. However, the location using the P picks from the geophone's z -component record (Figure 12c) yields a location result of $x = 5$ m, $y = 32$ m, and a depth of 15.5 m. Although the geophone P picks provide the most accurate result for the source depth, there is significant smearing of misfit values along the depth dimension that is observed neither on the DAS S picks nor on the geophone S picks. Therefore, the incidentally correct depth estimation is not particularly reliable.

DISCUSSION

We confirm in two distinct geologic settings that the main seismic phases are similarly recorded in DAS and the inline horizontal component of a collocated 3C geophone array. Recorded DAS data can be independently used in a variety of near-surface applications, but DAS does suffer from several limitations that need to be taken into account. Moreover, the geologic settings strongly influence the similarity between DAS and geophone measurements, especially in low-velocity environments.

A well-studied property of DAS measurements is the effect of the gauge length. We show that DAS data are strongly impacted by the gauge length in certain near-surface conditions. The limiting factor is the wavenumber k which is equal to the frequency f divided by

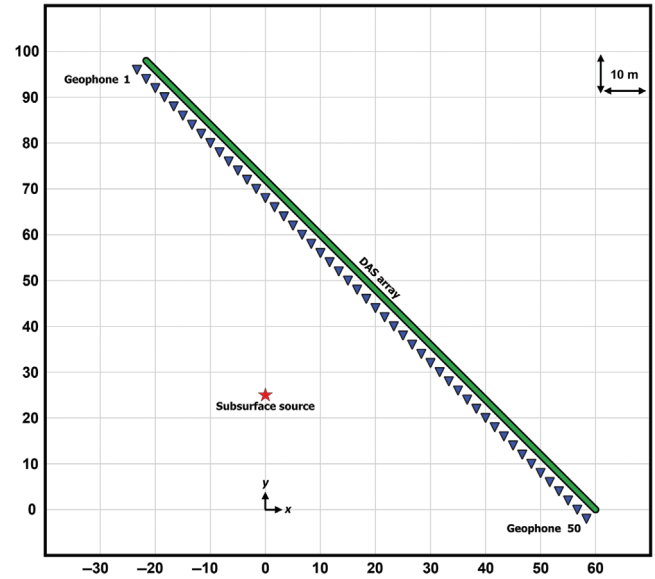


Figure 9. A map view of the acquisition setup. The DAS array is denoted by the green line. The geophones are indicated by the blue triangles and spaced by 2 m. Both are deployed over 98 m. The subsurface source is denoted by the red star, which is located at $x = 0$ m, $y = 25$ m, and a depth of 15 m.

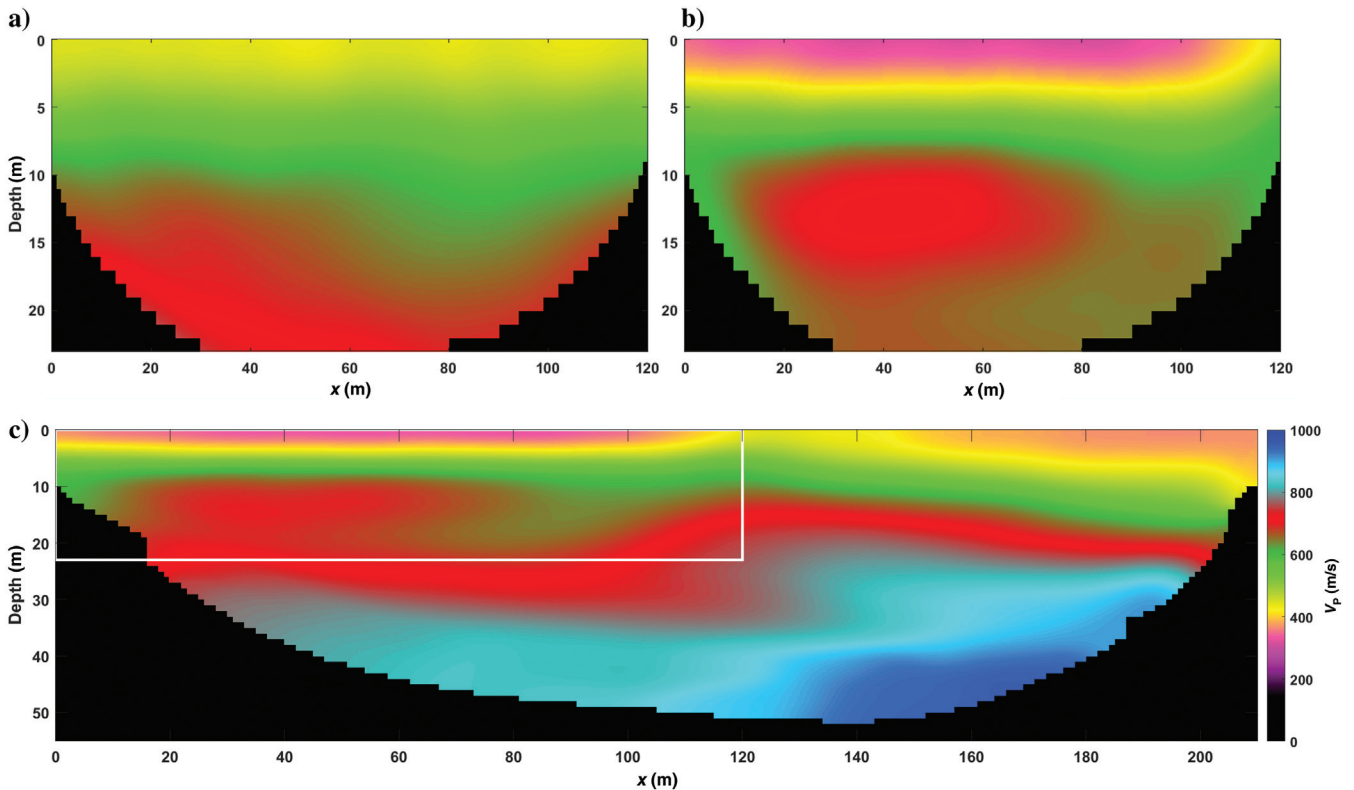


Figure 8. The P-wave velocity models obtained by the tomography. Decreased-size model from the 120 shots line recorded by (a) a 98 m long segment of the complete DAS array and (b) the 98 m long 3C geophone spread. (c) Combined model from the 120 shots line and the 210 m long DAS array and 98 m long 3C geophone spread. The white rectangle represents the dimensions of the reduced-size models. The black regions in the model represent areas with no coverage.

the phase velocity v . When the value of k approaches $1/GL$, data become unusable. Thus, generally speaking, high frequencies are more afflicted by the gauge length compared with low frequencies. For surface wave analysis, in which the phase velocity is low, the effect of the gauge length is much more prominent than when recording body waves. High-frequency low-velocity surface waves are most impacted by the gauge length and cannot be used during surface wave inversion. As a consequence, the ultrashallow structure cannot be accurately recovered. As a mitigation step, we show that a relatively simple analysis that only depends on the known gauge length can guide us to the reliable portion of the dispersion image and help us to avoid spurious events. For diving wave tomography, accurate traveltimes are essential. The averaging nature of the gauge length can potentially limit picking accuracy. In the ultrashallow subsection, the spatial averaging is larger than the depth of the target structure and thus strongly limits the resolution of any inversion approach.

DAS systems provide a directional strain measurement along the axis of the fiber. For P waves, the DAS response is thus proportional to $\cos^2 \theta$, where θ is the angle of incidence relative to the axis of the fiber. As a result, the ability of horizontally deployed DAS arrays to record upgoing P waves is limited, especially in the near offsets. At long offsets, depending on the velocity structure and hence on the angle of incidence upon reaching the DAS array, upgoing P waves may be recorded with some sensitivity. For impinging S waves and surface waves, DAS sensitivity to horizontal strain or strain rate allows for near-optimal recording. Near-offset traces are thus of limited use in diving wave tomography as it is based on upgoing P waves. In addition, the directionality of DAS measurements hinders passive location capabilities as, practically, only S waves can be reliably picked. As a consequence, the strongest constraint on event distance, the difference between the S- and P-arrival times, cannot be estimated.

A third limitation that we consider is the near-offset information. As we operate close to the source, usually within a few meters with a standard DAS setup, large strain values and, as a consequence, optical phase changes are recorded by the DAS system. Because the inter-

rogator operates as an optical interferometer, phase changes larger than π or smaller than $-\pi$ will be misinterpreted. The maximal recoverable strain is a function of the system design and optical parameters, mostly sensitivity, internal sampling rate, and gauge length. The internal sampling rate should be set to the highest possible frequency allowed by the system and the total length of the fiber. In other words, recording over a short fiber segment may prevent saturation. A shorter gauge length is preferable but comes at the expense of a lower signal-to-noise ratio (S/N). With our acquisition setup, which had a fixed 10 m gauge length over several kilometers of fiber, recorded data at short offsets are often unusable. Practically, this reduces

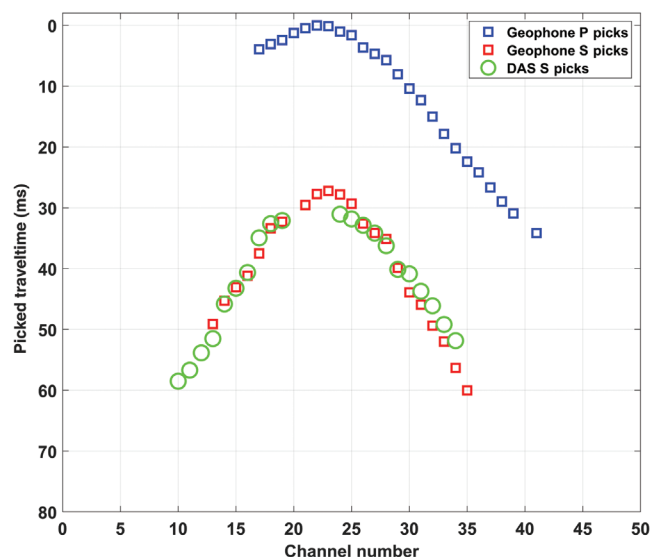


Figure 11. Picked traveltimes in each station for the DAS and geophones. The traveltimes are relative to the first arrival of the P wave that was picked on the geophone's z -component record. The geophone S picks are averaged picks from the x - and y -components, and the P picks are from the geophone's z -component only. Channels with no picks are not displayed.

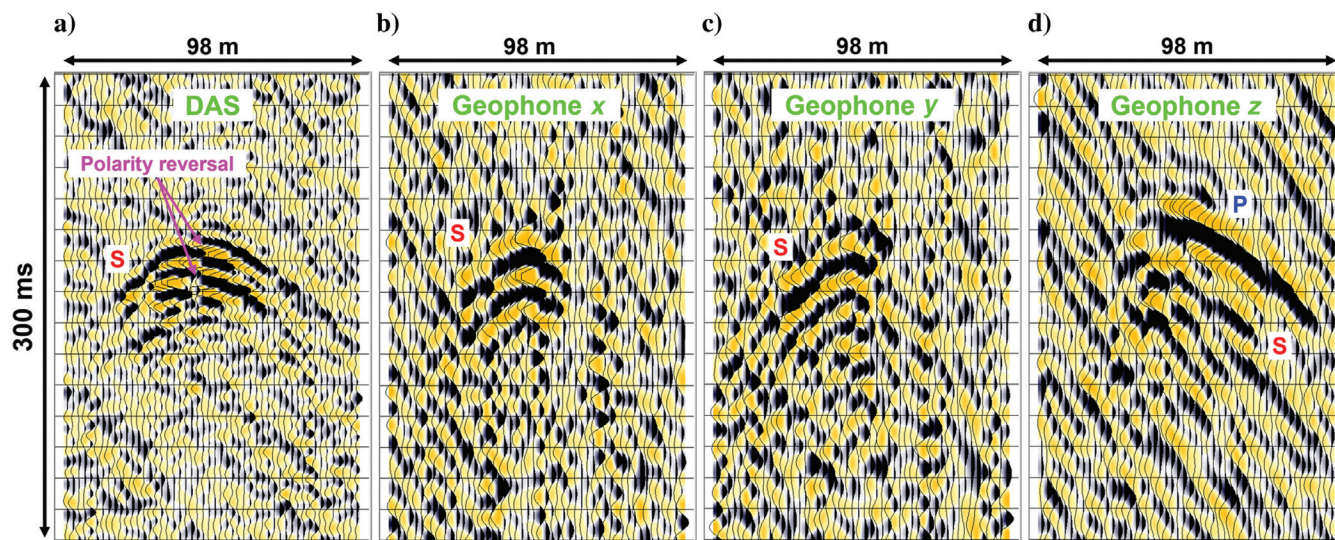


Figure 10. Shots from a subsurface source used for location: (a) DAS, (b) geophone's x -component, (c) geophone's y -component, and (d) geophone's z -component. First arrivals of P and S waves as well as examples of a polarity reversal are marked.

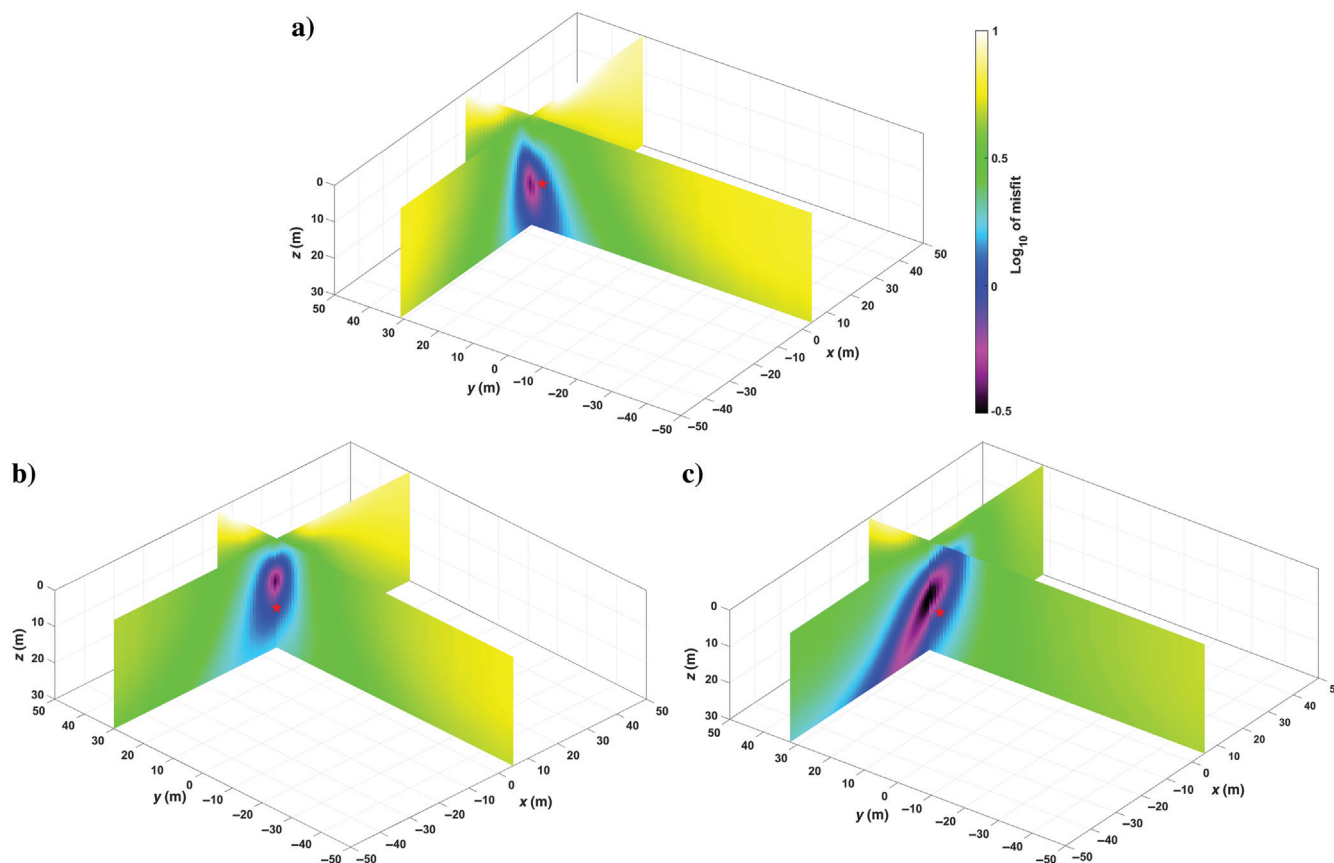


Figure 12. Comparison of DAS and geophone location results: (a) DAS using S picks, (b) geophone's x - and y -components using S picks, and (c) geophone's z -component using P picks. The true location of the subsurface source is indicated by the red star.

the resolution of diving wave tomography in the ultrashallow subsection. The depth below which the seismic velocity can be reliably recovered is a function of the optical setup and the subsurface structure and is in the range of several meters in this study.

However, DAS systems potentially hold key benefits for shallow seismic applications that are not highlighted in this study. Passive location results show that the source depth is not well resolved even when using geophones due to insufficient receiver coverage in the vertical axis, whereas the horizontal location is much better constrained. This indicates that the array geometry is the limiting factor in location accuracy. Downhole DAS deployments could potentially offer a much better depth constraint for a reasonable cost. Due to the directional measurement of DAS, polarity flips in recorded downhole data can directly indicate source depth after correction for the propagation path or with the assumption of mild velocity variations. Downhole DAS also can locally resolve the directivity issue for upgoing P waves, facilitating a vertical seismic profile-type analysis. Therefore, we are of the opinion that combining horizontal and downhole DAS deployments could significantly improve subsurface imaging and passive source location.

CONCLUSION

We confirm the similarity between DAS and collocated inline horizontal geophone measurements, shown in previous studies. DAS data can be used independently for surface wave analysis,

diving wave tomography, and passive source location, yielding reasonable results despite being inferior to geophone data. DAS is especially problematic in recovering the ultrashallow structure. Its directional measurement, dependence on the gauge length, and saturation at near offsets limit its usefulness.

To mitigate these limitations, we first suggest using the shortest possible gauge length, especially in low-velocity areas which accentuate the effect of the gauge length. There is a tradeoff between the gauge length and S/N, but we believe that, for many near-surface applications, a shorter gauge length with a lower S/N would be preferable. The internal sampling rate of the interrogator should be maximized to avoid optical phase saturation, thus commanding recording over short fiber sections where possible. Nonetheless, one should take into account that near-offset DAS channels might still be unusable, depending on the optical setup, source strength, and source-channel distance. We also can avoid using gauge-length-contaminated portions of the data by setting a simple frequency-wavenumber criterion. Applications that are based on upgoing P-wave energy, such as diving wave tomography, should consider the velocity structure and only use DAS channels for which noticeably nonvertical incidence is expected.

ACKNOWLEDGMENTS

We thank O. Shamir and G. Tzarfati for their hard work in the field and L. Abukrat for his helpful comments. In addition, we wish

to thank the editors, M. Mendoza, and two anonymous reviewers for their excellent critiques, comments, and suggestions that greatly improved the paper.

DATA AND MATERIALS AVAILABILITY

Data associated with this research are confidential and cannot be released.

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Biographies and photographs of the authors are not available.